AsmFEA - Getting started

Stress analysis of a whole SOLIDWORKS assembly, from a task pane, in a few clicks.

1 Find the pane | 2 Prepare | 3 Loads & fixtures | 4 Connections | 5 Run | 6 Results

1 Find AsmFEA in SOLIDWORKS

1. Start SOLIDWORKS (2018 or later; one of the versions you ticked in the installer). AsmFEA loads with it.
2. On the right of the window, the task pane has a new tab with the coloured hexagon icon (the one at the top of this page): click it to open AsmFEA.
3. If it is missing: **Tools > Add-ins**, tick **AsmFEA** in both columns, then OK.

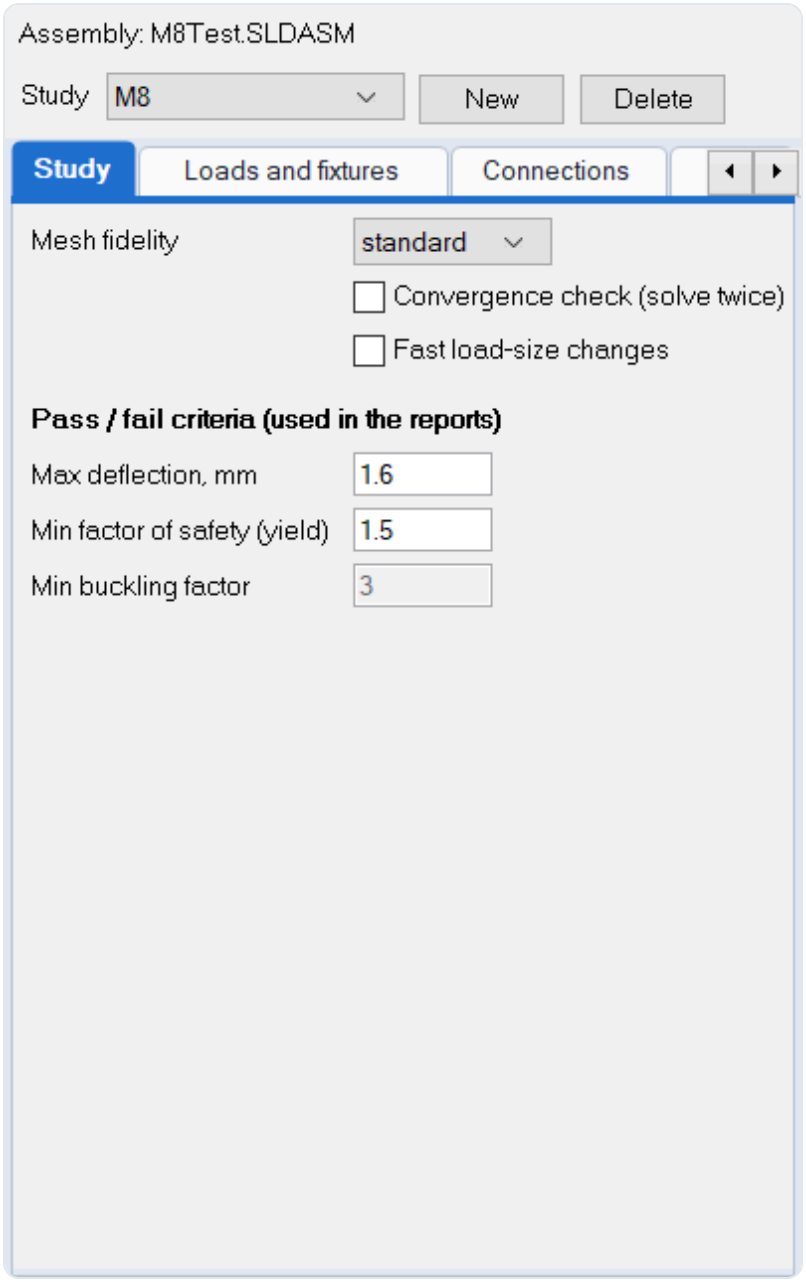
2 Prepare the assembly and start a study

- Open the assembly and give every part a material (right-click the part > **Material**). A single part works too: its bodies are treated as the parts, and bodies that touch are joined.
- Click New in the pane and name the study. It is saved inside the assembly (or part).
- On the **Study** tab choose what to **Optimise for**: *Speed* (a coarse mesh, one solve: quick, for comparing designs), *Balanced* (for most work) or *Accuracy* (a finer mesh for local stresses, and parts whose results have not settled are solved again on a finer mesh). AsmFEA decides the rest from this and your computer's free memory and cores, always leaving room for SolidWorks and other programs; the **Run** tab shows the plan before you run. *Advanced* lets you set the mesh level, convergence check, *Refine where not converged*

or solver yourself. Check the pass/fail criteria the reports use too. Changes are kept as you make them.

- If you expect to try several sizes of load, tick *Fast load-size changes*: each load is solved once, and changing only load sizes afterwards needs no new solve.

> Parts that touch are joined automatically. Mates tell AsmFEA where parts can slide or turn; weld beads are used too, and a sub-assembly is treated as one rigid piece.



The Study tab

3 Add fixtures and loads

1. Select one or more faces on the model.
2. On the **Loads and fixtures** tab click `Fixed` (held still) or `Pinned` (can rotate).
3. Loads belong to a **load case** (a set of loads solved together). Pick one at the top of the tab, or click `New load case`; every case is solved in the same run. Fixtures apply to all cases. Tick **Include buckling in load case** to check a case for buckling too (runs take longer).
4. Select faces and click `Force` or `Pressure`, or click `Gravity` for self-weight. Give the force as X, Y and Z components; the resultant shows as you type.
5. For a force acting away from the faces (a lever arm, a torque), also give how far from the faces' centre it acts, in X, Y and Z (leave 0, 0, 0 for a force on the faces).
6. For fluid in a tank, select the wetted faces and click `Hydrostatic`. Give the fluid's density (water 1000 kg/m^3), the absolute pressure on its top (1 atmosphere, 0.101325 MPa, for an open tank; the air outside balances it), and where the top of the fluid is, in X, Y and Z from the faces' centre; a ball and triad marks it on the model. The pressure rises with depth below it; faces above it are not loaded.

Click a row in the list to highlight its faces on the model; a load also shows an arrow pointing at where it acts. Double-click a row to change it.

Assembly: M8Test.SLDASM

Study M8 New Delete

Study | Loads and fixtures | Connections

Load case LC1

Buckling 5 modes

New load case | Rename | Delete load case

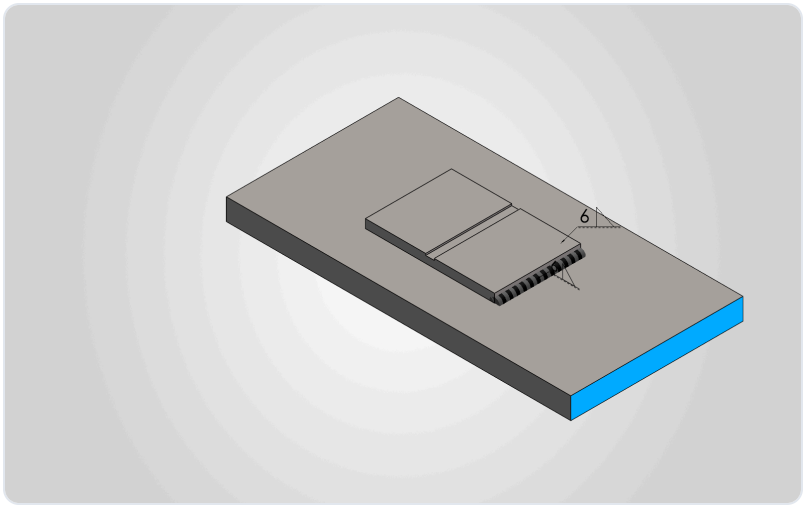
From the selected faces, add:

Fixed | Pinned | Force | Pressure

Gravity | Delete

Id	Kind	Type	Value	Faces
FX1	fixture	fixed		L3
LD2	load	gravity	0, -9810, 0 mm/s² (9810 mm/s²)	
LD1	load	force	0, -1000, 0 N (1000 N)	L1

Loads and fixtures

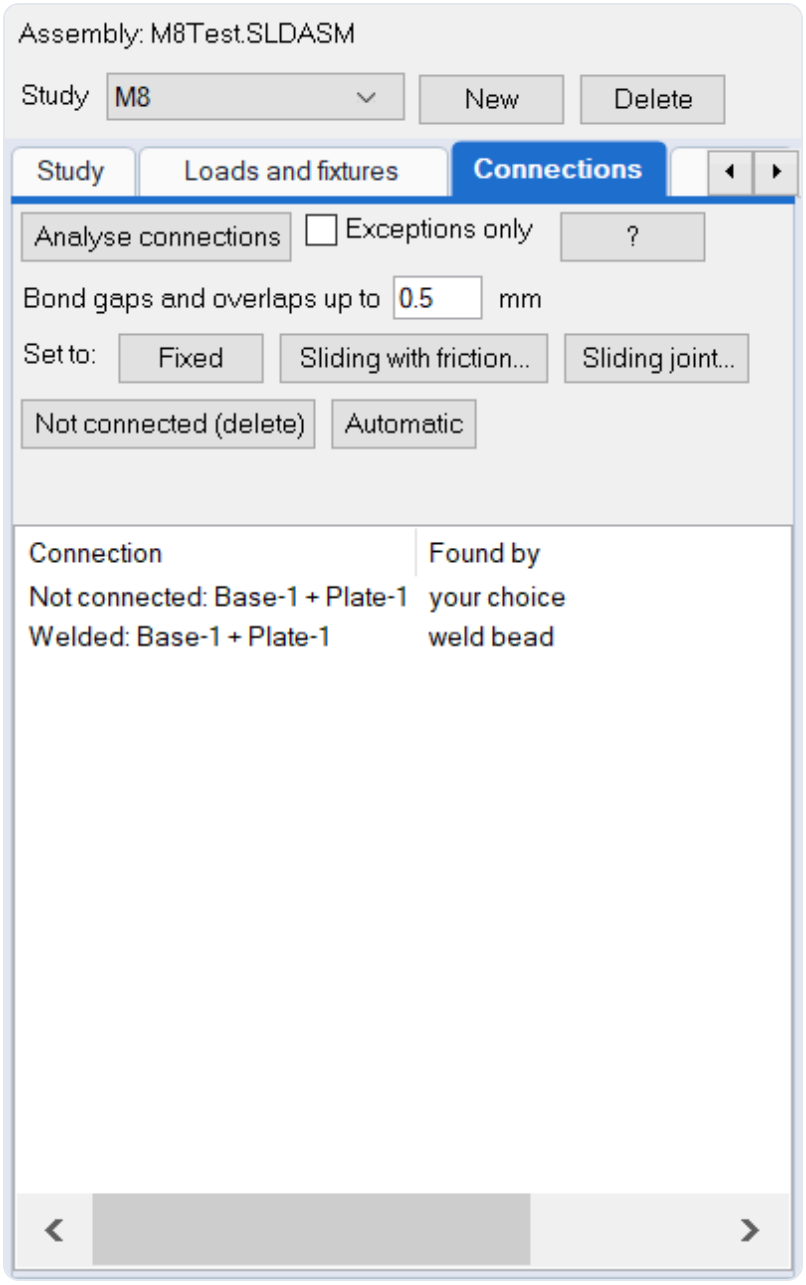


The fixed face highlighted in blue. This assembly is a plate on a base, held by a 6 mm weld bead along one edge.

4 Check how the parts connect

- Click `Analyse connections`: the list shows every place where parts meet and how they will be joined.
- To change one, select it and click `Edit...`, or use *Set to*: `Fixed` (bonded), `Sliding with friction...`, `Sliding joint...` or `Not connected`. `Automatic` undoes your choice.
- Weld beads appear as their own rows.

> Hover over any button for a tip. Double-click a connection or a warning, or press `?`, for a full explanation.

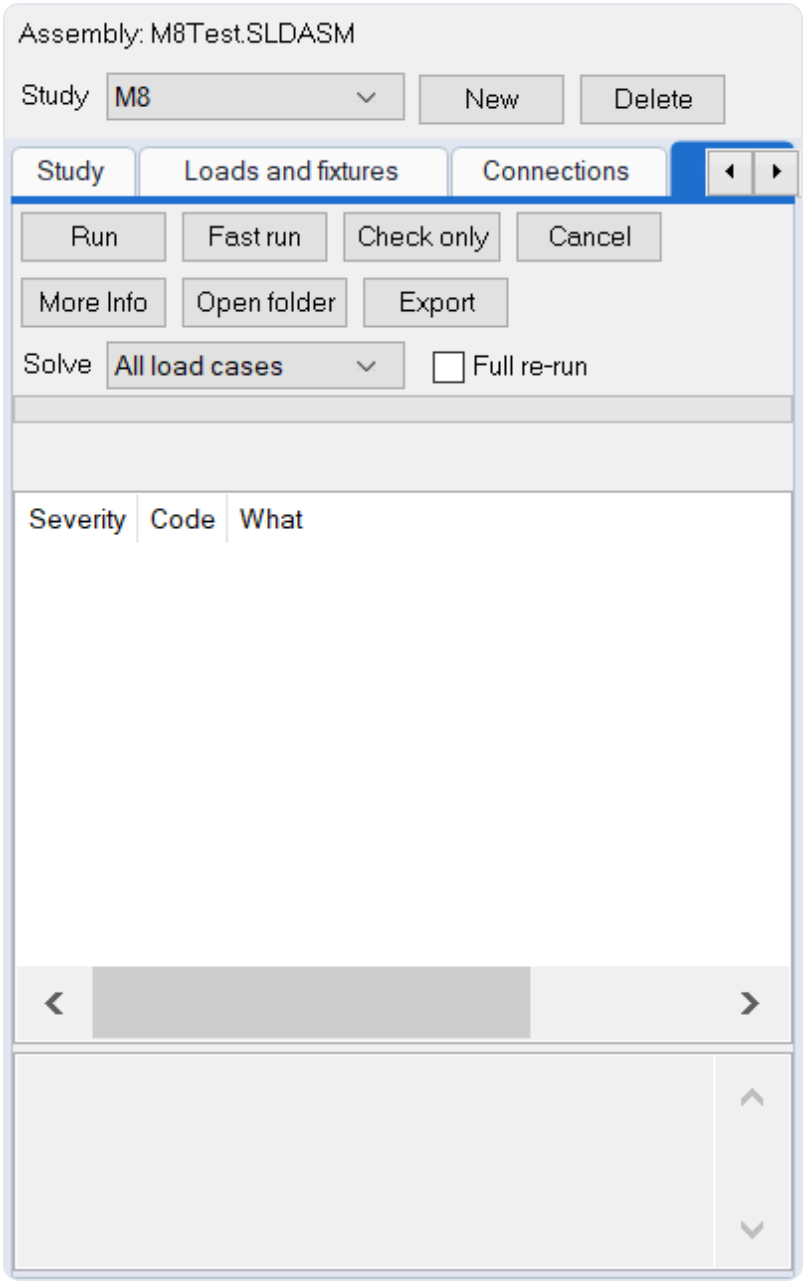


Connections, including a weld bead

5 Run

- On the **Run** tab click `Run`. The bar and the line below it show what is happening; a first run takes from under a minute to a few minutes.
- `Check only` just checks the set-up, quickly, without solving.
- `Fast run` is for trying design changes: a coarser model that shows where the problem is and whether a change helped, in a fraction of the time. Its results are marked **APPROXIMATE** and it writes no report: do a full `Run` at the end, before you rely on the numbers.

- A re-run redoes only what your changes need: with nothing changed it finishes in seconds, and a new pass/fail limit skips the solve. Tick *Full re-run* to redo everything.
- A model too large for CalculiX to fit in your computer's memory is no longer refused: AsmFEA solves it with its own hybrid solver, part by part, using the memory it has free. A note on the Run tab and in the report says when that happened.
- A run is never stopped for memory. If other programs take memory while it runs, a box says the solve is continuing more slowly; closing programs you don't need speeds it up again.
- The results say how far to trust them: a coloured line on the **Results** tab, and the reports, grade them *High*, *Good* or *Indicative* with the likely error on displacements and stresses, from how the model was solved. Click the line to see why.
- Anything that needs attention is listed: double-click it to see what it means and what to do.

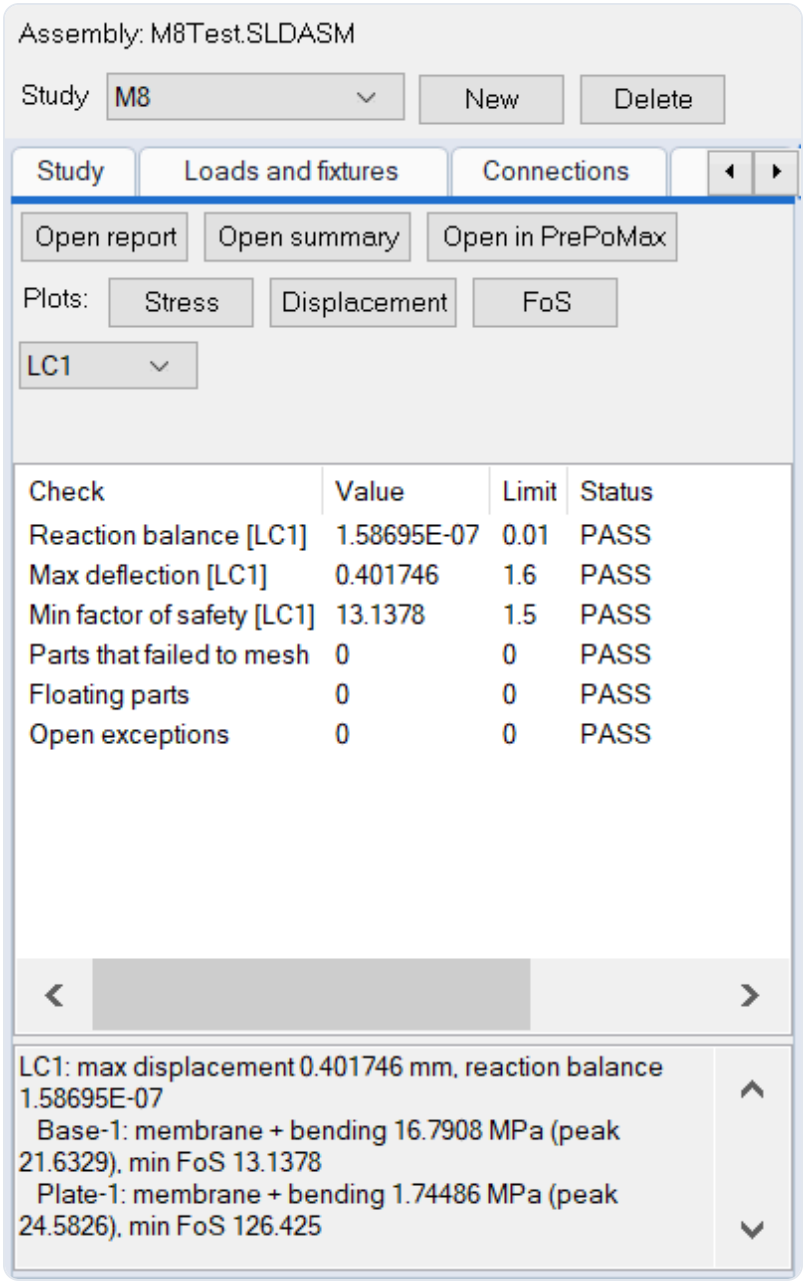


The Run tab

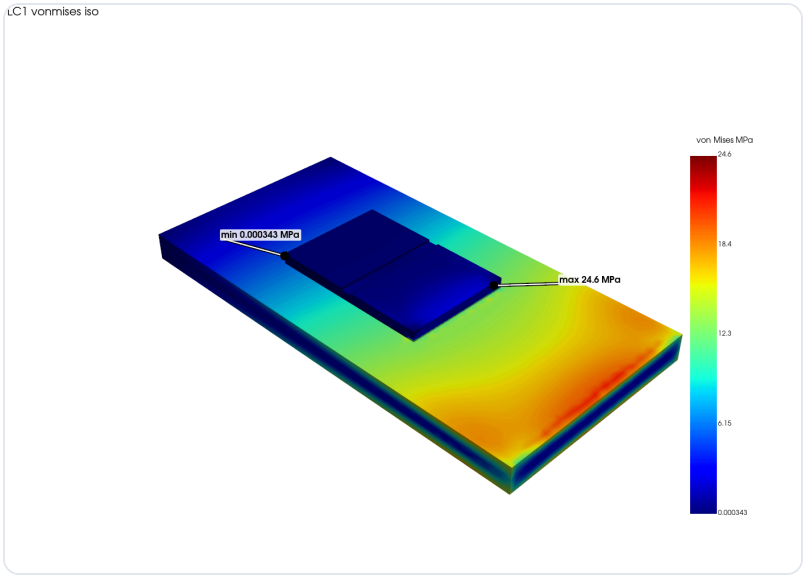
6 Look at the results

- The **Results** tab shows the pass/fail checks and the key numbers for each load case. Stress is judged as an engineer would: through the wall (membrane plus bending), round joints as well, and not on the spikes at sharp edges. The factor of safety is given on yield and, for information, on ultimate strength.
- `Stress`, `Displacement` and `FoS` open a 3-D window (their tooltips repeat what follows):

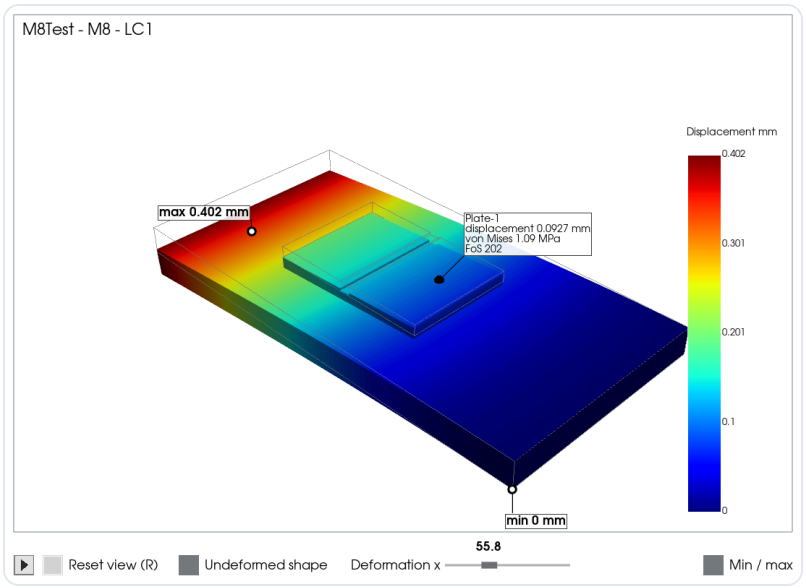
    - left drag to rotate, wheel to zoom, right drag to pan, as in SOLIDWORKS; **Reset view** (or R) goes back;
    - **click the model** to see the part and its stress, displacement and FoS at that point; click the background to clear it;
    - ▶ **play** ramps the load up and down so you can watch the stress build; click it again to stop at the full load;
    - **Deformation x** exaggerates the deformed shape, on every plot (0 shows it undeformed), and **Undeformed shape** adds a grey outline of the original;
    - **Min / max** shows the highest and lowest values; drag their labels anywhere;
    - **right-click a part** to see it as an outline and look at the parts behind it; right-click it again, or the background, to bring it back;
    - local peaks beyond the assessed stress show **magenta**; grey parts (such as bushes) are not assessed: check their bearing pressure by hand.
- `Open report` gives the full Word report; `Open summary` a half-page summary. Both need a full run (they are greyed out after a fast run).
- The pictures follow the way you look at the model in SOLIDWORKS: turn it, then open the results again.


Assembly: M8Test.SLDASM
Study
M8
New
Delete
Study
Loads and fixtures
Connections
Open report
Open summary
Open in PrePoMax
Plots:
Stress
Displacement
FoS
LC1
Check
Value
Limit
Status
Reaction balance [LC1]
1.58695E-07
0.01
PASS
Max deflection [LC1]
0.401746
1.6
PASS
Min factor of safety [LC1]
13.1378
1.5
PASS
Parts that failed to mesh
0
0
PASS
Floating parts
0
0
PASS
Open exceptions
0
0
PASS
LC1: max displacement 0.401746 mm, reaction balance 1.58695E-07
Base-1: membrane + bending 16.7908 MPa (peak 21.6329), min FoS 13.1378
Plate-1: membrane + bending 1.74486 MPa (peak 24.5826), min FoS 126.425

Results



Stress (von Mises), with the highest and lowest values marked



The 3-D window: displacement with the undeformed outline, a clicked point and the min / max labels

7 Optional: PrePoMax, for a closer look

PrePoMax is a free program for exploring results in detail: section cuts, values at any point, animations. AsmFEA does not need it, but `Open in PrePoMax` on the Results tab opens the last run in it.

1. Download it from prepomax.fs.um.si/downloads (version 2.6.0 is the one AsmFEA was tested with).

2. It is a zip file: unzip it anywhere, for example `C:\PrePoMax`. There is nothing to install.
3. In AsmFEA click `Open in PrePoMax` and choose **Find PrePoMax.exe...** the first time; AsmFEA remembers it.

✓ How accurate is AsmFEA?

The verification report (also in the Start menu, and as a PDF in the AsmFEA folder) compares AsmFEA with classical hand calculations on four test cases (a cantilever, a beam under its own weight, two members joined end to end, and a buckling column) and checks the CalculiX solver against its own test examples and two NAFEMS benchmarks. Every comparison came within its acceptance criterion, the largest difference being 0.6 %.

Satisfy yourself of its accuracy for your own work by running your own tests against hand calculations or other trusted results before relying on it.

8 Keep your work

- The study is saved **inside the assembly**: save the assembly to keep it.
- Results and reports are in a folder next to the assembly called `<assembly name>_FEA`.
- Units are millimetres, newtons and MPa throughout.

> Always check its results with engineering judgement and a hand calculation before relying on them for anything that matters.

AsmFEA uses CalculiX (solver) and Gmsh (mesher), free software downloaded by the installer; see the licenses folder.